A TO Z MASTER GUIDE

Tableau A to Z

Data Connection - Calculations - LOD - Dashboards - Interview Questions

Prepared specially for Koushik
Beginner-friendly A to Z edition

Inside: WORKSHEETS | CHARTS | FILTERS | LOD | ACTIONS | PROJECT

How to use	Oka topic chadivi code ni own-ga type cheyyi. First output guess cheyyi, tarvatha run cheyyi. Interview answer boxes ni loud-ga 2 times practice cheyyi.

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How to Use This Guide

Ee PDF Tableau ni zero nunchi interview level varaku nerchukovadaniki. Prathi main topic lo step, reason, expected result, common mistake and interview answer untayi. Menu names software version batti konchem change avvachu, kani workflow same untundi.

Step	What you do	Why
1. Observe	Concept and example chaduvu	Meaning ardham avutundi
2. Predict	Result mundu guess cheyyi	Logic build avutundi
3. Perform	Own-ga steps repeat cheyyi	Tool memory vastundi
4. Explain	Result enduku vachindo cheppu	Interview clarity vastundi
5. Practice	Two new questions solve cheyyi	Independent skill build avutundi

Interview answer formula	Definition one line -> process 2 or 3 points -> small example -> business use -> limitation. Ee order lo answer chepthe neat-ga and confident-ga untundi.

1. Tableau Foundation

1.1 What is Tableau?

Tableau is a visual analytics platform used to connect to data, explore it through interactive visualizations, build dashboards and publish insights. Tableau Desktop/Public are authoring experiences; Tableau Cloud/Server support managed sharing and collaboration.

Easy Telugu	Fields ni Rows, Columns and Marks card meeda drag chesthe Tableau visual query create chesi marks draw chestundi. Manam data question batti fields and aggregation select chestam.

1.2 Product overview

Product	Use / note
Tableau Desktop	Full authoring and local publishing workflow
Tableau Public	Free public portfolio publishing; content is public
Tableau Cloud	Hosted analytics platform
Tableau Server	Self-managed enterprise deployment
Tableau Prep	Visual data preparation and flow building

Privacy	Confidential/company/personal data Tableau Public ki publish cheyyaku. Public portfolio kosam sample or anonymized data use cheyyi.

2. Connect to Data

2.1 First connection

Step	Action	Result
1	Connect pane -> Microsoft Excel/Text/Server	Choose source
2	Data Source page -> drag/select tables	Logical model
3	Verify field names and data types	Correct analysis behavior
4	Choose Live or Extract where applicable	Connection mode
5	Open Sheet 1	Start analysis

2.2 Live vs extract

Live	Extract
Queries source during interaction	Stores optimized snapshot
Freshness depends on source	Fast portable analytics in many cases
Performance depends on database/network	Requires refresh to update

Choice	Fast source and real-time need unte Live. Performance/offline/portable snapshot need unte Extract. Data volume, refresh and governance batti decide cheyyi.

3. Relationships, Joins and Unions

3.1 Relationship vs physical join

Relationship	Join
Logical tables remain separate	Rows combined into one physical table
Context-based joins at analysis time	Fixed join type/clauses
Preserves each table's level of detail	Can duplicate or filter rows based on cardinality
Recommended in many multi-table cases	Useful when row-level combination is required

3.2 Create relationship

Data Source canvas logical layer lo second table drag cheyyi -> relationship noodle select -> matching fields choose -> cardinality/referential settings only if known.

3.3 Join

Double-click logical table to physical layer -> drag second table -> select inner/left/right/full and join clauses. Preview row counts and unmatched keys.

3.4 Union

Physical layer lo New Union or drag compatible tables below one another. Same columns monthly files stack avutayi; missing fields NULL values create chestayi.

Why totals duplicate?	One order many return rows/join matches unte sales repeat avvachu. Relationship often original grain preserve chestundi; join use chesthe cardinality and row count validate cheyyi.

4. Dimensions, Measures, Discrete and Continuous

Concept	Meaning	Visual effect
Dimension	Categorical field used to slice/group	Headers/labels/marks
Measure	Numeric field commonly aggregated	Axis or encoded magnitude
Discrete (blue)	Finite headers/categories	Headers
Continuous (green)	Continuous range	Axis

4.1 Example

Category to Columns and SUM(Sales) to Rows -> one bar per category. Category partitions rows; Sales is summed inside each partition.

Blue vs green	Color field data type kaadu; discrete vs continuous role ni indicate chestundi. Date ni discrete Year headers or continuous timeline ga use cheyyachu.

4.2 Aggregation

Measures default aggregation SUM/AVG etc. Pill dropdown -> Measure -> choose aggregation. Analysis > Aggregate Measures toggle disaggregate cheyyagaladu, but purpose understand chesi use cheyyi.

5. Worksheet Anatomy

Area	Purpose
Data pane	Fields, folders, sets, parameters
Columns shelf	Horizontal structure/axis
Rows shelf	Vertical structure/axis
Marks card	Color, Size, Label, Detail, Tooltip, Shape
Filters shelf	Limit data in view
Pages shelf	Step through a field
Show Me	Suggests visual types based on selected fields

5.1 Build and read marks

Each mark represents a combination of dimensions in the view at a particular level of detail. Adding Customer Name to Detail can multiply marks even when label is hidden.

Why number of marks increased?	New dimension Marks/Rows/Columns lo add chesthe view granularity finer avutundi. Tableau more groups create chesi more marks draw chestundi.

6. Core Charts - Steps and Reasons

6.1 Bar chart

Step	Action	Why
1	Category to Rows	One header per category
2	Sales to Columns	Bar length encodes SUM(Sales)
3	Sort descending	Ranking readable
4	Sales to Label if few bars	Exact values
5	Format currency/units	Business readability

6.2 Line chart

Order Date to Columns as continuous Month/Exact Date, Sales to Rows. Continuous date creates time axis. If Month date part only is discrete, same month across years may combine depending on fields.

6.3 Scatter plot

Sales to Columns, Profit to Rows, Customer/Product to Detail, Category to Color. Each entity becomes a mark; reference lines can create quadrants.

6.4 Map

Assign geographic role to State/City, double-click field, add Sales to Size and Profit to Color. Ambiguous locations need Country/Region context or Edit Locations.

6.5 Heatmap/highlight table

Region to Rows, Category to Columns, Sales to Color and Label. Color shows magnitude; use diverging palette only around meaningful center such as zero.

6.6 Pie/donut caution

Use only for very few part-to-whole categories. Sorted bars compare accurately when categories are many or values close.

7. Filters and Order of Operations

7.1 Filter types

Filter	Typical stage
Extract	Limits extract data
Data source	Limits all sheets using source
Context	Creates context subset for dependent filters/logic
Dimension	Filters categorical members
Measure	Filters aggregated values
Table calculation	Filters post-visual table calculation results

7.2 Top N with context

Customer filter Top 10 by Sales, then Region normal dimension filter pedithe Top 10 may be global first. Region ni Add to Context chesthe region subset first, then Top 10 within region calculate avvachu.

Result enduku change?	Tableau filters fixed sequence lo evaluate avutayi. Context filter earlier stage lo subset create chestundi; downstream Top N/conditional filters aa subset meeda work chestayi.

Context caution	Every filter context cheyyaku. Specific order/performance need kosam use cheyyi and result validate cheyyi.

8. Calculated Fields

8.1 Row-level calculation

```
IF [Sales] > 0 THEN [Profit] / [Sales] END
```

Each row margin calculate chesi view aggregation apply avvachu. Business overall margin kosam ratio of sums often correct.

8.2 Aggregate calculation

```
SUM([Profit]) / SUM([Sales])
```

Why values differ?	AVG(row margin) each row/customer equal weight ivvachu. SUM(Profit)/SUM(Sales) sales-weighted overall margin. Business definition decide cheyyi.

8.3 IF, CASE and NULL

```
IF ISNULL([Region]) THEN "Unknown"
ELSEIF [Profit] < 0 THEN "Loss"
ELSE "Profit"
END
```

8.4 Aggregate/non-aggregate error

`IF [Category] = ... THEN SUM([Sales]) END` mixes row-level Category and aggregate Sales. Aggregate Category using ATTR/MIN where semantically valid, or make calculation row-level then aggregate.

9. Table Calculations

9.1 What is a table calculation?

A table calculation transforms values already present in the visualization. It depends on addressing/partitioning and the current view layout.

```
Running total: RUNNING_SUM(SUM([Sales]))
Percent of total: SUM([Sales]) / TOTAL(SUM([Sales]))
Previous difference: SUM([Sales]) - LOOKUP(SUM([Sales]), -1)
```

9.2 Compute Using

Pill dropdown -> Edit Table Calculation. Table Across, Down, Pane options or Specific Dimensions define direction. Wrong compute using is the most common reason for wrong totals.

How to debug	View lo row/column headers observe cheyyi. Calculation which dimension along move avvali, which dimensions partition/restart cheyyali ani decide cheyyi.

10. Level of Detail Expressions

10.1 FIXED

```
{ FIXED [Customer Name] : SUM([Sales]) }
```

View lo other dimensions unna lekapoyina customer level sales calculate chestundi, subject to Tableau order of operations and relevant filters.

10.2 INCLUDE and EXCLUDE

```
{ INCLUDE [Customer Name] : SUM([Sales]) }
{ EXCLUDE [Sub-Category] : SUM([Sales]) }
```

LOD	Meaning	Use
FIXED	Specified dimensions, independent of view detail	Customer lifetime sales
INCLUDE	Adds finer dimension to view grain	Average customer sales by region
EXCLUDE	Removes a view dimension	Category total while showing sub-category

FIXED filter behavior	FIXED dimension filter mundu evaluate avvachu. Filter LOD ni affect cheyyali ante context filter or different LOD/design use cheyyi; order of operations understand cheyyi.

11. Parameters, Sets, Groups, Bins and Hierarchies

11.1 Parameter for metric switch

Step	Action
1	Data pane -> Create Parameter -> String list Sales, Profit, Orders
2	Show Parameter
3	Create calculated field using CASE [Metric]
4	Place calculation on visual
5	Use parameter in dynamic title

```
CASE [Metric]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
WHEN "Orders" THEN COUNTD([Order ID])
END
```

11.2 Sets

Set defines members in/out based on manual selection, condition or Top N. Use to compare Top customers vs Others. Set actions can update membership through dashboard interaction.

11.3 Groups, bins and hierarchy

- Group combines selected dimension members, such as spelling variants.
- Bin divides continuous numeric values into intervals, useful for histograms.
- Hierarchy organizes drill path such as Country > State > City.

12. Dashboard Building

12.1 Create dashboard

Step	Action	Reason
1	New Dashboard icon	Open canvas
2	Set fixed/range/automatic size	Target device
3	Drag horizontal/vertical containers	Grid layout
4	Drag sheets into containers	Add views
5	Add filters/legends/text/navigation	Interaction/context
6	Use Layout pane padding/borders	Consistent spacing
7	Test fit, filters and device preview	Usability

12.2 Tiled vs floating

Tiled objects snap into layout and resize predictably. Floating objects allow exact overlay but can be harder across screen sizes. Prefer containers for main structure.

Blank space issue	Container hierarchy, fixed size, padding and sheet Fit setting inspect cheyyi. Random floating positions kanna tiled containers stable.

13. Dashboard Actions and Interactivity

Action	Use
Filter action	Selected marks filter target sheets
Highlight action	Emphasize related marks
Go to URL	Open contextual external link
Go to Sheet	Navigate to another sheet/dashboard/story
Change Parameter	Update parameter from selected mark
Change Set Values	Update a set interactively

13.1 Create filter action

Dashboard > Actions > Add Action > Filter -> source/target sheets -> Run on Select/Hover/Menu -> clearing selection behavior. Test with multiple marks and empty selection.

13.2 Use as Filter

Dashboard sheet menu funnel icon is quick filter action. For precise behavior use Dashboard > Actions settings.

Unexpected clearing	Clearing selection options Keep filtered values, Show all values, Exclude all values result ni control chestayi. Requirement batti select cheyyi.

14. Tooltips, Reference Lines and Forecasting

14.1 Tooltip

Marks > Tooltip edit chesi only useful fields and context add cheyyi. Viz in Tooltip supported scenarios lo another sheet embed cheyyachu; performance consider cheyyi.

14.2 Analytics pane

- Reference line: target, average or benchmark.
- Trend line: statistical relationship; interpret assumptions carefully.
- Forecast: supported time series; inspect model and prediction intervals.
- Cluster: exploratory grouping; not automatic business truth.

15. Performance and Publishing

15.1 Performance checklist

- Reduce data to needed rows/columns and use efficient extracts where suitable.
- Prefer relationships and correct grain over row-exploding joins.
- Limit high-cardinality marks and unnecessary quick filters.
- Use context filters only for justified order/performance cases.
- Simplify expensive calculations and test database performance.
- Use Performance Recording to inspect query/layout/render time.
- Hide unused fields and optimize extract refresh strategy.

15.2 Publish

Server > Publish Workbook for authorized Cloud/Server, choose project, data source authentication, permissions and refresh. Tableau Public uses Save to Tableau Public and makes workbook/data publicly accessible subject to platform behavior.

16. Complete Superstore Dashboard Project

16.1 Requirements

- Metric parameter: Sales, Profit, Orders.
- KPIs: selected metric, profit margin, return rate.
- Trend by month, category breakdown, map, top customers/products.
- Year/Region filters and show/hide filter panel.
- Drill or action from overview to detail.

16.2 KPI calculations

```
Profit Margin
SUM([Profit]) / SUM([Sales])

Order Count
COUNTD([Order ID])

Metric Value
CASE [Metric]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
WHEN "Orders" THEN COUNTD([Order ID])
END
```

16.3 Build sequence

Phase	Action	Validation
Connect	Orders plus Returns/People relationships	Key and grain
Fields	Types, geographic roles, hierarchy	Date/map behavior
Calculations	Metric, KPIs, LOD/table calcs	Text table checks
Worksheets	Build one question per sheet	Titles and sorting
Dashboard	Containers, KPIs, filters, actions	Interaction test
Publish	Public/Cloud with safe data	Refresh and permissions

Presentation flow	Overview dashboard first answer: performance enti? Trend: eppudu? Breakdown/map: ekkada? Detail/action: evaru/edi drive chestundi? Recommendation: next action enti?

17. Tableau Interview Questions with Model Answers

Q: 1. What is Tableau?

Answer: A visual analytics platform for connecting to data, exploring it through interactive visualizations and publishing dashboards.

Q: 2. Dimension vs measure?

Answer: Dimensions categorize/slice data; measures are values commonly aggregated for analysis.

Q: 3. Discrete vs continuous?

Answer: Discrete fields create headers and distinct categories; continuous fields create axes across a range.

Q: 4. Blue vs green pill?

Answer: Blue indicates discrete, green continuous; it is about the field's role in the view, not simply its data type.

Q: 5. Live vs extract?

Answer: Live queries the source during analysis. An extract is an optimized snapshot that needs refresh.

Q: 6. Relationship vs join?

Answer: Relationships keep logical tables separate and combine contextually. Joins combine rows physically with a fixed join definition.

Q: 7. Union?

Answer: Stacks rows from compatible tables, aligning columns by field names.

Q: 8. Data blending?

Answer: Combines aggregated results from separate data sources using linking fields; relationships/cross-database options are often preferred when available.

Q: 9. What is a mark?

Answer: A visual representation of one data point/group at the current level of detail.

Q: 10. Marks card?

Answer: Controls visual encodings such as Color, Size, Label, Detail, Tooltip and Shape.

Q: 11. Why marks increase?

Answer: Adding a dimension refines view granularity, creating more unique dimension combinations.

Q: 12. Filter types?

Answer: Extract, data source, context, dimension, measure and table calculation filters execute at different stages.

Q: 13. Context filter?

Answer: Creates a primary subset used by dependent filters and some calculations; use for correct order or measured performance needs.

Q: 14. Order of operations?

Answer: The sequence Tableau uses to evaluate filter and calculation types, which explains why similar filters can produce different results.

Q: 15. Calculated field?

Answer: A reusable expression deriving a new value from existing fields, functions, parameters and calculations.

Q: 16. Aggregate vs row-level calculation?

Answer: Row-level computes per underlying row before aggregation. Aggregate calculations operate on summarized values.

Q: 17. Table calculation?

Answer: A post-query calculation over values in the current visualization, controlled by addressing and partitioning.

Q: 18. Compute Using?

Answer: Defines the direction/dimensions across which a table calculation runs and where it restarts.

Q: 19. LOD expression?

Answer: A calculation controlling aggregation granularity independently or relative to the dimensions in the view.

Q: 20. FIXED vs INCLUDE vs EXCLUDE?

Answer: FIXED uses specified dimensions; INCLUDE adds detail; EXCLUDE removes selected view detail.

Q: 21. Why FIXED ignores a filter?

Answer: FIXED may be evaluated before normal dimension filters. Make the filter context or redesign based on required order.

Q: 22. Parameter?

Answer: A single user-controlled value independent of field membership, used in calculations, filters, reference lines and actions.

Q: 23. Set?

Answer: A dynamic or fixed subset of dimension members divided into In/Out.

Q: 24. Group vs set?

Answer: A group combines members into categories. A set defines in/out membership and supports Top/condition/action logic.

Q: 25. Bin?

Answer: A generated dimension dividing a continuous measure into equal-sized intervals.

Q: 26. Hierarchy?

Answer: An ordered drill path of related dimensions, such as Country to State to City.

Q: 27. Dual axis?

Answer: Overlays two measures with separate axes; synchronize only when scales and interpretation justify it.

Q: 28. Dashboard vs story?

Answer: A dashboard combines views in one interactive canvas. A story sequences sheets/dashboards as narrative points.

Q: 29. Tiled vs floating?

Answer: Tiled objects participate in layout containers; floating objects have independent positions and can overlay.

Q: 30. Dashboard action?

Answer: An interaction triggered by hover/select/menu that filters, highlights, navigates, opens URLs or changes parameters/sets.

Q: 31. Set action?

Answer: Updates set membership based on user interaction with marks.

Q: 32. Parameter action?

Answer: Updates a parameter from a selected mark or summary value.

Q: 33. Show Me?

Answer: Suggests visual types compatible with selected fields; analytical judgment is still required.

Q: 34. ATTR?

Answer: Returns a value when all underlying rows share it, otherwise displays an asterisk, useful for detecting mixed values.

Q: 35. COUNT vs COUNTD?

Answer: COUNT counts non-null records/values; COUNTD counts distinct values.

Q: 36. Why totals duplicate after join?

Answer: A one-to-many or many-to-many physical join repeats values. Verify grain, keys and relationship alternatives.

Q: 37. How to improve dashboard performance?

Answer: Reduce data/marks/filters, use suitable extracts, efficient calculations, fewer sheets and inspect Performance Recording.

Q: 38. How to create dynamic metric selector?

Answer: Create a parameter, CASE calculation returning the selected measure, show parameter and use dynamic title.

Q: 39. How to build Top N by region?

Answer: Use Top N set/filter and make Region a context filter when business logic requires Top N inside selected region.

Q: 40. How do you validate a Tableau dashboard?

Answer: Reconcile totals, inspect grain/joins, test filters/actions/empty selections, check labels/sorts and compare known cases.

Q: 41. How do you explain a Tableau project?

Answer: State business question, data connection, model choices, calculations, dashboard design, actions, validation, publishing and insights.

18. Practical Interview Tasks

Task	Strong approach
Build Sales vs Profit dashboard	Question -> grain -> sheets -> containers -> actions -> validate
Top 10 changes incorrectly	Explain order of operations and context filter
Wrong table calculation	Edit Compute Using and partition/address fields
Join duplicates sales	Profile key cardinality and use relationship/pre-aggregation
Dynamic metric	Parameter + CASE + title/tooltips
Slow dashboard	Performance recording, marks, filters, extracts, calculations

19. 30-Day Tableau Plan

Days	Topics	Output
1-3	Connect, interface, fields	First workbook
4-7	Bar/line/map/scatter/heatmap	Chart practice
8-10	Filters, groups, bins, hierarchy	Interactive sheets
11-13	Calculated fields	KPI library
14-16	Table calculations	Running/rank/percent views
17-19	LOD expressions	Customer/category analysis
20-21	Parameters and sets	Metric/Top N controls
22-24	Dashboard containers/design	Overview dashboard
25	Actions/tooltips/navigation	Interactive UX
26	Performance/publish	Publishing checklist
27-28	Superstore project	Portfolio dashboard
29-30	Interview/mock	40 answers + timed build

Official References and Version Note

Software interfaces update avutayi. Ee guide concepts and click paths official documentation tho verify chesanu. Option exact place kanipinchakapothe ribbon search/help use cheyyi.

Official source	URL
Tableau Desktop help	https://help.tableau.com/current/pro/desktop/en-us/default.htm
Relationships and joins	https://help.tableau.com/current/pro/desktop/en-us/joining_tables.htm
LOD expressions	https://help.tableau.com/current/pro/desktop/en-us/calculations_calculatedfields_lod.htm
Order of operations	https://help.tableau.com/current/pro/desktop/en-us/order_of_operations.htm
Table calculations	https://help.tableau.com/current/pro/desktop/en-us/calculations_tablecalculations.htm
Create dashboards	https://help.tableau.com/current/pro/desktop/en-us/dashboards_create.htm

Official source	URL
Dashboard actions	https://help.tableau.com/current/pro/desktop/en-us/actions_dashboards.htm